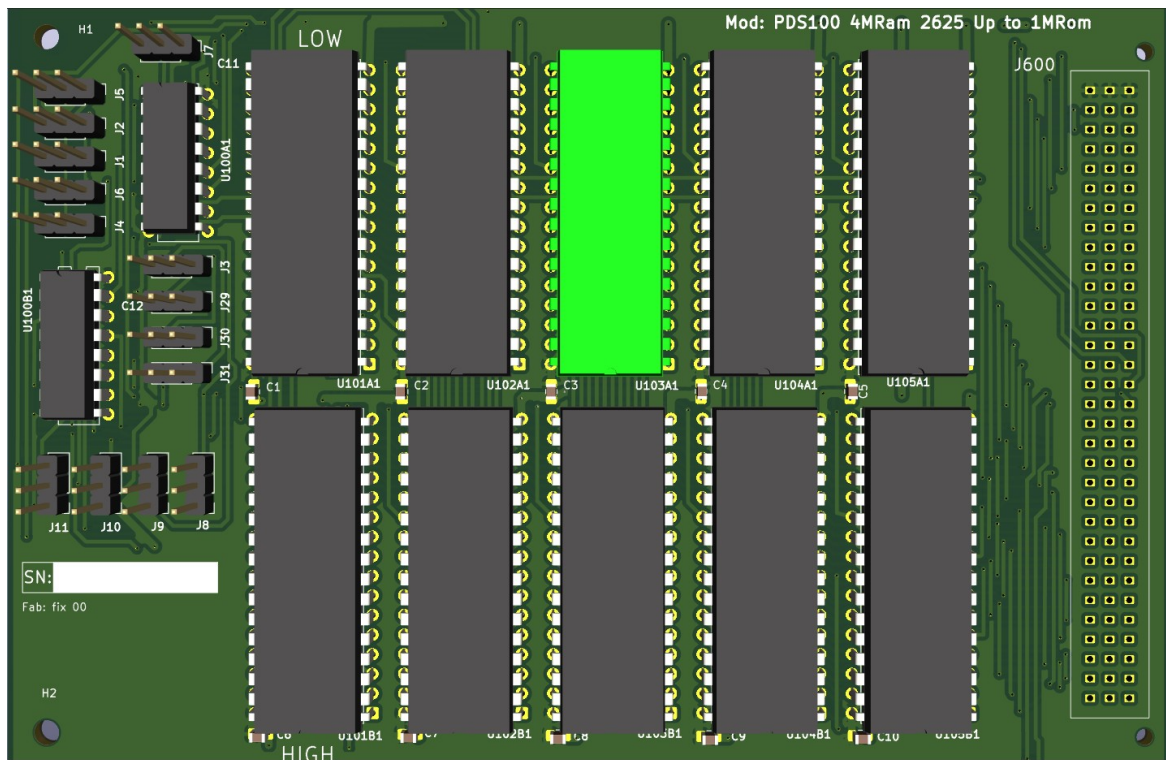


SISTEM PDS317
Technical Module Manual



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1. General Module Specifications

1.1. Total Capacity and Bus Organization (16-bit)

The **PDS317** system architecture supports a maximum addressable workspace of **8 MBytes (4 MWords)** of Static RAM. In the current **Revision 1.0** hardware release, the memory board is partially populated with **1 MByte** of active RAM.

For non-volatile firmware storage, the system decodes a dedicated **1 MByte** address space for ROM. The initial **Revision 1.0** assembly utilizes a **64 KByte (32 KWords)** ROM chip, mapped appropriately for system initialization and boot vectors.

Orion68K Memory Map (Initial Release)

Address Range (Hex)	Size (Words)	Resource Type	Population Status
\$000000 – \$07FFFF	512 KW	ROM Space	Populated (64 KB

Address Range (Hex)	Size (Words)	Resource Type	Population Status
			Active)
\$080000 – \$0FFFFFF	512 KW	RAM Bank 0	Populated (1 MB Active)
\$100000 – \$17FFFF	512 KW	RAM Bank 1	Unpopulated
\$180000 – \$1FFFFFF	512 KW	RAM Bank 2	Unpopulated
\$200000 – \$27FFFF	512 KW	RAM Bank 3	Unpopulated
\$280000 – \$2FFFFFF	512 KW	RAM Bank 4	Unpopulated
\$300000 – \$37FFFF	512 KW	RAM Bank 5	Unpopulated
\$380000 – \$3FFFFFF	512 KW	RAM Bank 6	Unpopulated
\$400000 – \$47FFFF	512 KW	RAM Bank 7	Unpopulated
\$480000 – \$4FFFFFF	512 KW	RAM Bank 8	Unpopulated
\$500000 – \$57FFFF	512 KW	RAM Bank 9	Unpopulated
\$580000 – \$5FFFFFF	512 KW	RAM Bank 10	Unpopulated

1.2 Operating Voltages and Estimated Current Consumption

Component Specifications

SRAM (Static RAM)

- **Component:** AS6C4008 (Maximum of 8 units)
- **Voltage Range:** 2.7V to 5.5V (Nominal: 5V)
- **Speed:** 55 ns
- **Operating Temperature:** -40°C to +85°C
- **Current Consumption (per unit):** 30 mA (Max)

EEPROM (ROM)

- **Component:** AT28C256 (Maximum of 2 units)
- **Voltage Range:** 5V ± 10%
- **Speed:** 150 ns
- **Operating Temperature:** -55°C to +125°C
- **Current Consumption (per unit):** 50 mA (Max)

Estimated Maximum Power Consumption

A tabela abaixo apresenta o cálculo do consumo estimado da placa populada em sua capacidade máxima de memória.

Component Type	Part Number	Unit Current	Max Qty	Total Current (Max)
SRAM	AS6C4008	30 mA	8	240 mA
EEPROM	AT28C256	50 mA	2	100 mA
Total (Memory Subsystem)	—	—	—	340 mA

Note: O consumo total acima reflete apenas o subsistema de memória operando em capacidade máxima. Certifique-se de que a linha de +5V da fonte de alimentação do sistema

possui uma margem de segurança de pelo menos 20% acima do consumo total da placa somado aos demais periféricos (como CPU e controladores de I/O).

2. Electrical Architecture and Bus Distribution

2.1. Memory Bank Organization: LOW Bank and HIGH Bank

Organizing memory into **LOW Bank** and **HIGH Bank** is a classic and elegant engineering solution. It is most commonly used in **16-bit architectures** (such as the Motorola 68000 or Intel 8086) to allow a 16-bit CPU to talk to 8-bit memory chips efficiently, or in **banking/paging systems** to overcome address space limitations.

Since the PDS317 uses 8-bit chips (the AS6C4008 is a 512K x 8 SRAM and the AT28C256 is a 32K x 8 EEPROM), dividing them into **LOW** and **HIGH** banks is the standard way to achieve a wider data path or structural memory segmentation.

Here is a professional, technical breakdown of this topic in English, tailored for your hardware manual:

Memory Bank Organization: LOW Bank and HIGH Bank

In computer architecture, organizing memory into **LOW Bank** and **HIGH Bank** is a technique used to align physical memory chips with the data bus width of the processor, or to segment memory spaces for specific architectural requirements.

For a system utilizing 8-bit memory chips (such as the AS6C4008 SRAM and AT28C256 EEPROM), this organization typically serves one of two primary purposes: **16-Bit Interleaved Access** or **Address Space Paging**.

2.2 16-Bit Bus Alignment (Interleaved Organization)

If the system CPU features a **16-bit data bus** but utilizes **8-bit wide memory chips**, the memory must be split into two parallel banks to allow the CPU to read or write a full 16-bit word in a single clock cycle.

- **LOW Bank (Even Bank):**

- Connects directly to the lower half of the data bus (**\$D_0 - D_7\$**).
- Contains all data located at **even memory addresses** (\$0x00000\$, \$0x00002\$, \$0x00004\$, etc.).
- Activated when the CPU asserts the **Lower Data Strobe (LDS)** or when the lowest address line (\$A_0\$) is 0.

- **HIGH Bank (Odd Bank):**

- Connects directly to the upper half of the data bus (**\$D_8 - D_{15}\$**).
- Contains all data located at **odd memory addresses** (\$0x00001\$, \$0x00003\$, \$0x00005\$, etc.).
- Activated when the CPU asserts the **Upper Data Strobe (UDS)** or when a specific control line dictates an odd-byte access.

How It Works in Practice:

- **Byte Access (8-bit):** If the CPU only needs 1 byte, it activates either the LOW bank or the HIGH bank depending on whether the address is even or odd.
- **Word Access (16-bit):** If the CPU needs a full 16-bit word, it activates **both banks simultaneously**. The LOW bank provides the lower byte, and the HIGH bank provides the upper byte, merging into a single 16-bit transfer.

2.3 Segmented / Paged Memory Organization (8-Bit Systems)

If the PDS317 is a pure **8-bit system** (where both the CPU and the chips are 8-bit), "LOW Bank" and "HIGH Bank" refer to **address space segmentation**—often used to separate different types of data or to implement Bank Switching to bypass CPU addressing limits.

- **LOW Bank (Base / System Memory):**

- Usually mapped to the lower part of the CPU’s address space (e.g., \$0000_h to \$7FFF_h\$).
- Typically holds critical, non-volatile code such as the system ROM/BIOS, vector tables, or primary system RAM that must always be accessible.
- **HIGH Bank (Swappable / Extended Memory):**
 - Mapped to the upper part of the address space (e.g., \$8000_h to \$FFFF_h\$).
 - Often controlled by logic gates or a latch (Bank Switching). By toggling a control register, the CPU can "swap" different physical chips into this single logical space, allowing a CPU with a 64KB limit to access megabytes of RAM.

Application to the PDS317

Given that your board supports a maximum of **8 RAM chips** and **2 ROM chips**:

Bank Designation	Hardware Mapping	Typical Function
LOW Bank	First set of SRAM/ROM chips	Handles Even Bytes (16-bit mode) OR maps to Lower Address Space (8-bit mode).
HIGH Bank	Second set of SRAM/ROM chips	Handles Odd Bytes (16-bit mode) OR maps to Upper Address Space/Paged Expansion (8-bit mode).

 Engineering Note

When designing the PCB layout for LOW and HIGH banks, trace length matching for the data lines (\$D_0-D_7\$ vs \$D_8-D_{15}\$) and control signals (Chip Select $\overline{CS}$, Output Enable $\overline{OE}$) is critical to prevent propagation

delays, ensuring both banks respond within the tight 55ns window of the AS6C4008 SRAM.

2.4 MC68000 Control Signals: /LDS, /UDS, and /AS

To handle the 16-bit architecture we just discussed (dividing memory into **LOW/Even** and **HIGH/Odd** banks), the Motorola 68000 uses three critical control signals: Address Strobe (/AS), Upper Data Strobe (/UDS), and Lower Data Strobe (/LDS).

In the 68k world, the forward slash (/) or an overbar indicates an **active-low** signal, meaning the line is considered "true" or "asserted" when its voltage drops to 0V (GND). Here is how these three signals work together to manage the bus and select your memory banks:

- **Address Strobe (/AS)**

The /AS signal is the CPU's way of saying: *"The address currently on the address bus (\$A_1 - A_{23}) is stable and valid. Go ahead and decode it."*

- **Why it matters:** Because address lines don't all change state at the exact same picosecond, they "glitch" briefly when changing from one address to another.
- **The Fix:** The 68000 waits until all address lines settle, then drops /AS to low. Your memory decoding logic (like a GAL or discrete gates) should **always** use /AS as an enabling input to prevent the memory chips from accidentally reacting to those brief address glitches.

- **Upper Data Strobe (/UDS) and Lower Data Strobe (/LDS)**

These two signals are the magic sauce that allows a 16-bit CPU to talk to 8-bit memory banks. Because the 68000 can read or write either a single 8-bit byte or a full 16-bit word, it needs a way to tell the memory chips exactly which byte(s) it wants.

Notice that the MC68000 does not have an \$A_0\$ (bit 0) address pin! Instead, /UDS and /LDS replace \$A_0\$ to select between even and odd bytes.

- **/UDS (Upper Data Strobe):** Automatically managed by the CPU to control the **HIGH Bank** (Data lines \$D_8 - D_{15}\$). It targets **even addresses**.
- **/LDS (Lower Data Strobe):** Automatically managed by the CPU to control the **LOW Bank** (Data lines \$D_0 - D_7\$). It targets **odd addresses**.

- **How They Route to Your PDS317 Memory Banks**

When connecting these to your 8-bit AS6C4008 SRAM or AT28C256 ROM chips, you use `/UDS` and `/LDS` to drive the Chip Select (`/CS`) or Output/Write Enable (`/OE`, `/WE`) lines of the respective banks.

Here is the truth table of how the 68000 uses these signals during a bus cycle:

<code>/AS</code>	<code>/UDS</code>	<code>/LDS</code>	Bus Operation Type	Active Memory Bank	Active Data Bus Lines
1	X	X	None (Address is invalid)	None (Chips disabled)	High Impedance (Tri-stated)
0	0	0	16-bit Word Access	Both HIGH and LOW Banks	Full Bus (<code>\$D_0 - D_{15}</code>)
0	0	1	8-bit Byte Access (Even Address)	HIGH Bank only	Upper Bus (<code>\$D_8 - D_{15}</code>)
0	1	0	8-bit Byte Access (Odd Address)	LOW Bank only	Lower Bus (<code>\$D_0 - D_{7}</code>)

• **Hardware Design Tip for your GAL Logic**

When writing your equations for memory chip selection, you must combine your address decoding with these strobes. For example:

- **HIGH Bank RAM Select:** Should only activate when the address matches your RAM range **AND** `/AS` is low **AND** `/UDS` is low.
- **LOW Bank RAM Select:** Should only activate when the address matches your RAM range **AND** `/AS` is low **AND** `/LDS` is low.

If you don't factor `/UDS` and `/LDS` into your chip select logic during a **Write** cycle, writing a single byte to an even address would accidentally overwrite the adjacent odd byte with bus garbage!

3. Layout Engineering and Signal Integrity

3.1. Power Rail Dimensioning (1 mm VCC and GND Traces)

Engineering Note: Unlike modern multi-layer PC motherboards that utilize dedicated internal power planes to distribute voltage with minimal resistance, the PDS100 RAM board is designed on a 2-layer PCB. In a 5V retrocomputing environment populated with high-density static RAM (SRAM) and logic chips, transient current spikes during simultaneous switching can cause severe ground bounce and VCC sagging if the traces are too narrow.

To mitigate this without the luxury of internal power planes, a deliberate design choice was made to use robust **1mm traces** for all primary VCC and GND routing. This significantly reduces DC resistance and trace inductance, acting as a low-impedance backbone across the entire 10-chip matrix. Combined with local decoupling, this heavy-duty copper layout guarantees stable voltage delivery, suppresses high-frequency power rail noise, and ensures the system operates like a tank even under heavy bus loads.

4. Backplane Interface and Configurations

4.1. DIN 41612 Connector Pinout (Memory Slot)

Architectural Advantages

The **DIN 41612 Eurocard connector standard** is widely recognized as the industry backbone for high-reliability modular systems, such as VMEbus, NuBus, and custom homebrew computer backplanes. For a 16-bit system like the MC68000, using a 3-row, 96-pin **DIN 41612 Type C** connector for the memory expansion slot provides critical mechanical and electrical advantages over standard pin headers.

Key Advantages of the DIN 41612 Design

- **Structural Robustness and Mechanical Alignment**

Unlike standard dual-row pin headers, the DIN 41612 features a robust shroud and explicit keying mechanism.

- **Anti-Warping:** The high-quality thermoplastic housing prevents the connector pins from bending during insertion or removal.

- **Mating Cycle Longevity:** Engineered for heavy-duty deployment, these connectors maintain reliable contact resistance over hundreds of insertion/extraction cycles, ensuring the modular cards don't develop intermittent connections over time.
- **Superior Signal Integrity and Crosstalk Mitigation**

High-speed address and data bus transitions on an MC68000 system can easily induce electromagnetic interference (EMI) and capacitive crosstalk between adjacent pins.

 - **Three-Row Shielding Architecture:** Having three parallel rows (A, B, and C) allows hardware designers to interleave Ground (GND) pins around high-frequency control signals (such as /AS, /UDS, /LDS, and clock lines). Row B can act as a massive structural shield layer between the active traces of Rows A and C.
- **Low-Impedance Power Delivery**

The DIN 41612 contact pins are rated for higher current density compared to dense IDC or edge connectors. By dedicating multiple parallel pins at the outer boundaries of the connector to +5V and GND, the backplane provides a very low-impedance power distribution network, drastically reducing voltage sag and high-frequency supply ripple when multiple SRAM chips enable simultaneously.
- **Memory Slot Pinout Architecture (16-Bit MC68000 Bus Optimized)**

The matrix below details an optimized 96-pin layout tailored specifically for a memory card expansion slot. It balances data bus symmetry, interleaves ground connections to prevent ringing, and establishes a massive power plane entry at the connector edges.

Pin #	Row A Signal	Row B Signal (Shielding/Aux)	Row C Signal
1	+5V (Main Power Input)	+5V (Main Power Input)	+5V (Main Power Input)
2	DO	GND	/RESET
3	D1	GND	RESET
4	D2	GND	IRQ7
5	D3	GND	/AS
6	D4	GND	/LDS
7	D5	GND	UDS
8	D6	GND	/IORD

Pin #	Row A Signal	Row B Signal (Shielding/Aux)	Row C Signal
9	D7	GND	/CPURD
10	D8	GND	/CPUWR
11	D9	GND	/IOWR
12	D10	GND	IRQ3
13	D11	GND	IRQ4
14	D12	GND	A23
15	D13	GND	A22
16	D14	GND	A21
17	D15	GND	A20
18	A0	GND	/IO_CS1
19	A1	GND	IRQ5
20	A2	GND	/IO_CS
21	A3	GND	/MEM_CS
22	A4	GND	A19
23	A5	GND	A18
24	A6	GND	A17
25	A7	GND	A16
26	A8	GND	A15
27	A9	GND	A14
28	A10	GND	FREE
29	A11	GND	IRQ1
30	A12	GND	-5V (Auxiliary Rail)
31	A13	GND	+12V (Auxiliary Rail)
32	GND (Main Return)	GND (Main Return)	GND (Main Return)

- **Implementation Notes for the Memory Subsystem Layout**
- **Symmetric Data Paths:** Spreading the data bus across Row A (D0–D7) and Row B (D8–D15) keeps the physical path lengths highly balanced when transitioning from the backplane onto the memory card, which minimizes differential propagation delay between the LOW and HIGH banks during 16-bit word cycles.
- **Pre-Decoded Chip Selects (/MEM_CS0,/MEM_CS1):** Bringing pre-decoded chip select lines from a master logic device (e.g., a system GAL located on the CPU or backplane supervisor

board) onto pins 23 and 24 minimizes the amount of discrete glue logic gates required directly on the memory board, streamlining the expansion hardware footprint.

4.2. Configuration Jumper Map (ROM/RAM Selection)

The PDS317 features a highly flexible hardware configuration memory map managed via onboard jumpers. This design serves two primary strategic purposes:

- **ROM Footprint Versatility:** It allows the two primary non-volatile sockets to accept either standard **28-pin** or **32-pin** memory packages. While this manual frequently references the 28-pin **AT28C256 EEPROM** as the default system baseline, the hardware natively supports 32-pin devices like the **AT29C512 Flash Memory**, which was utilized during the initial development phases of the PDS317.
- **Socket Reallocation (RAM Expansion):** Because the Orion68K ecosystem is modular and interconnected via a backplane, a dedicated CPU card might already carry its own surface-mount (PLCC) boot ROM. To prevent the PDS317's two ROM sockets from being wasted in such scenarios, these jumpers allow the user to reconfigure those sockets to accept standard **AS6C4008 SRAM** chips instead, significantly expanding the system's volatile

memory capacity.

Primary Memory Selection Jumpers (J3, J29, J30, J31)

These four jumpers must be configured as a matching set depending on the specific integrated circuit (IC) type and package installed in the ROM/Universal sockets.

Jumper J3

Position	Selection Mapping
1 - 2	ROM AT28C256 (28-pin) OR RAM AS6C4008
2 - 3	ROM AT29C512 (32-pin)

Jumper J29

Position	Selection Mapping
1 - 2	ROM AT28C256 (28-pin) OR RAM AS6C4008
2 - 3	ROM AT29C512 (32-pin)

Jumper J30

Position	Selection Mapping
1 - 2	ROM AT28C256 (28-pin)
2 - 3	ROM AT29C512 (32-pin) OR RAM AS6C4008

Jumper J31

Position	Selection Mapping
1 - 2	RAM AS6C4008
2 - 3	ROM AT29C512 (32-pin)

• Hardware Configuration & Static Signal Jumpers

These jumpers route specific static control lines, hardware configurations, or are tied to default logical states for the current revision.

• Jumpers J6, J7, J8, J10, and J11 (Default Closed 2-3)

These jumpers must remain closed on pins 2-3 for normal operation in revision 1.0.

Jumper	Position 1 - 2	Position 2 - 3
J6	N/A	CLOSED (Default)
J7	N/A	CLOSED (Default)
J8	N/A	CLOSED (Default)
J10	N/A	CLOSED (Default)
J11	N/A	CLOSED (Default)

Jumper J12 (Default Closed 1-2)

Position	Status
1 - 2	CLOSED (Default)

Position	Status
2 - 3	N/A

• **Reserved / Future Expansion Jumpers (J4, J9)**

 **Hardware Note:** These jumpers are allocated for future hardware expansion subroutines and experimental traces that have not yet been fully validated. For **PDS317 Hardware Version 1.0**, these **must be left completely open** (no jumper shunts installed).

Jumper J4

Position	Selection / Function
1 - 2	N/A (Leave Open)
2 - 3	N/A (Leave Open)

Jumper J9

Position	Selection / Function
1 - 2	N/A (Leave Open)
2 - 3	N/A (Leave Open)

5. Address Decoding Logic and Chip Select (/CS)

5.1. Address Decoding Subsystem (74HC138)

The memory bank selection logic on the board is managed by a pair of **74HC138 3-to-8 line decoders/demultiplexers** (U100A1 and U100B1). This circuit takes the high-order address lines from the CPU bus and decodes them into individual, active-low Chip Select (/CS) signals for the physical RAM and ROM/Universal memory chips.

The architecture is split into two identical decoder stages to natively support the 16-bit data bus structure of the MC68000:

- **U100A1 (LOW Bank Decoder):** Generates chip selects for the lower byte (_L) memory chips connected to data lines D_0 - D_7.

- **U100B1 (HIGH Bank Decoder):** Generates chip selects for the upper byte (_H) memory chips connected to data lines D_8 - D_15.

Active Decoding Logic (Validated Configuration)

In the current validated configuration, the system utilizes address lines **BA19, BA20, and BA21** mapped to the selection pins (A_0, A_1, A_2) of both decoders to slice the memory map into distinct blocks.

1. Enable Gates Integration (/MEM_CS, /LDS, /UDS)

For a decoder output to assert low, all three hardware enable pins (E0, E1, E2) must be satisfied simultaneously:

- **Primary Block Enable (E0):** Driven by /MEM_CS (a master pre-decoded memory space signal from the backplane/GAL). Both decoders are kept in standby until this signal drops low.
- **Data Strobe Masking (E1):** * On U100A1 (LOW Bank), this pin is tied to /LDS. It ensures the decoder only activates during lower-byte or full 16-bit word operations.
 - On U100B1 (HIGH Bank), this pin is tied to /UDS. It ensures the decoder only activates during upper-byte or full 16-bit word operations.
- **Signal Routing Output Map**

When enabled, the binary combination of the validated address lines BA19 A_0, BA20 A_1, BA21 A_2) routes the selection to the matching memory chips:

- **Y0 Output:** Generates /MEM1CS_L and /MEM1CS_H (typically used for the boot ROM sockets).
- **Y1 to Y4 Outputs:** Generate individual chip selects for the static RAM expansion array (/RAM1CS through /RAM4CS for both banks).

Hardware Jumpers for Future/Experimental Configuration

The layout features several 3-pin configuration headers (J2, J4, J6, J7, J8, J9, J10, J11) that allow the inputs of the 74HC138 chips to be altered.

 **Important:** The jumper selections involving static Ground, ROMRAM9900_CS, and address line BA22 are architectural provisions for future system updates. They are **not yet validated** for Revision 1.0.

Address Line Demultiplexing & Overrides (J6, J7, J8, J10, J11)

- **Validated State (Revision 1.0):** Shunts must be placed on pins **2-3** of these headers. This directly bypasses the experimental paths and safely routes BA19, BA20, and BA21 straight to the decoder inputs.
- **Future Provision (Pins 1-2):** Shunting pins 1-2 allows the address lines to be disconnected from the inputs, forcing those specific decoder pins to a static hardware **Ground (GND)** or alternate custom signals like ROMRAM9900_CS via J4 and J9. Upper Memory Space Expansion Header (J2)
- **Validated State (Revision 1.0):** This header manages the active-high enable input (E2, pin 6) of both decoders. In the baseline version, it is tied to a configuration that keeps the decoders permanently unblocked during standard memory cycles.
- **Future Provision (BA22):** Traces are available on pin 3 to route the higher-order CPU address line **BA22** directly into the E2 gate. When populated in future configurations, this will automatically double the decodable address space by disabling or enabling the entire 74HC138 network depending on whether BA22 is high or low.

6. Diagnostics and Bench Testing

6.1. Essential On-Board Test Points (TP)

In the board revision A does not have test points, they are missing.

6.2. 68k Assembly Routine for Memory March Test (RAM Test)

```

; =====
; PDS100 SRAM MARCH TEST ROUTINE (16-BIT WORDS)
; =====
                ORG      $00001000      ; Example execution address in RAM/ROM

RAM_START      EQU      $00002000      ; Define your PDS100 RAM base address here
RAM_END        EQU      $003FFFFF      ; Define your 4MB upper limit here
TEST_PAT1      EQU      $5555          ; Alternating bit pattern 1 (0101...)
TEST_PAT2      EQU      $AAAA          ; Alternating bit pattern 2 (1010...)

START_TEST:
    LEA        RAM_START,A0            ; Load start address into Pointer A0
    MOVE.L     #TEST_PAT1,D0           ; Load first test pattern into D0

; --- WRITE PHASE 1 ---
WRITE_LOOP:
    MOVE.W     D0,(A0)+                 ; Write pattern to current address, move to next
    CMPA.L     #RAM_END,A0             ; Check if we reached the end of RAM
    BNE        WRITE_LOOP              ; If not, keep writing

; --- READ & VERIFY PHASE 2 ---
    LEA        RAM_START,A0            ; Reset pointer to start of RAM
VERIFY_LOOP:
    MOVE.W     (A0)+,D1                 ; Read value from RAM into D1
    CMP.W      D0,D1                   ; Compare read value with original pattern
    BNE        RAM_FAIL                ; If mismatch, jump to fail routine
    CMPA.L     #RAM_END,A0             ; Check if we verified all memory
    BNE        VERIFY_LOOP             ; If not, keep verifying

RAM_PASS:
    ; Put your "LED On" or Serial Out Success code here (System Alive!)
    BRA        RAM_PASS

RAM_FAIL:
    ; Address of failure is stored in A0 (minus 2 due to post-increment)
    ; Put your Error Handling / Halt code here
    ILLEGAL                                ; Force exception for debugging

```

Date	Revision	Author	Mod.
2026/06/22	A	pdsilva	Initial

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